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August 24, 2026 · 5:00 – 7:30 PM · Kai Santos

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MECH 191 — Metallurgy · Week 1 Notes · fill in blanks & key terms as you go

Topics

- What is metallurgy — and why does it matter?
- The atom: protons, neutrons, electrons, valence shells
- The Periodic Table — groups, periods, and metals
- Atomic bonding — metallic, ionic, covalent
- Alloys and solid solutions
- Crystal structures — SC, BCC, FCC, HCP
- Grains and grain boundaries

Reference Material

Kalpakjian 6th — Ch. 3
Structure and Manufacturing
Properties of Metals

- Crystal Structures Visualization
- Periodic Table of Elements
- Grain Boundaries and Types

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The science of metals — understanding structure, properties, and behavior so we can choose, test, treat, and work with them effectively.

Copper: abundant, conductive — but weak. Adding ~10-20% Tin disrupts copper's crystal lattice, creating Bronze: significantly harder and stronger than either metal alone, while keeping copper's workability and corrosion resistance. The first alloy in human history.

Pure metals rarely provide the properties we need. Alloys let us engineer the material for the job.

Adding a second element disrupts the crystal structure, blocking deformation and changing properties.

Carbon in iron → steel. Chromium in steel → stainless. Every unit in this course builds on this idea.

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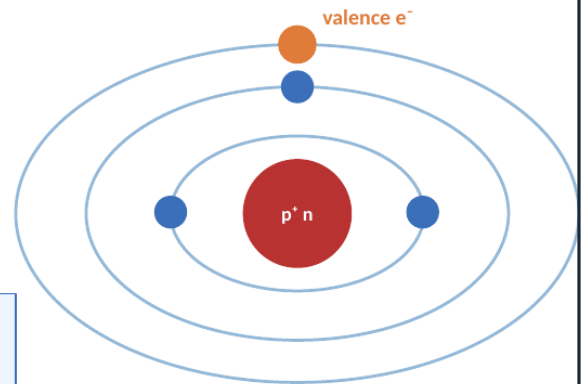
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Three key components:

Positive charge — inside the nucleus — defines the element (atomic number)

No charge — inside the nucleus — defines mass. Isotopes vary in neutron count.

Negative charge — orbit in energy bands/shells around the nucleus



The outermost electrons — the only ones involved in bonding. Carbon has 4 valence electrons, bonding in 4 directions simultaneously. This is why adding carbon to iron creates steel — carbon atoms sit between iron atoms and block deformation.

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The Periodic Table

Periods (Rows)

Organized by electron shells. Moving down adds shells — each period is a new shell level.

Groups (Columns)

Organized by valence electrons. Same group = similar bonding behavior and chemical properties.

Our Focus: Groups 3-12

Transition metals — Fe, Cu, Ni, Cr, Ti, Mo, and all engineering metals live here.

Classification by the staircase line:

Metals

Left of the staircase

- Electrically & thermally conductive
- Malleable and ductile
- Lustrous (shiny surface)
- Solid at room temp (except Hg)

Metalloids

The staircase itself

- Properties depend on conditions
- Semiconducting — critical for electronics
- Behavior: part metal, part non-metal
- Si, Ge, As, Sb, Te, Po, At

Non-metals

Right of the staircase

- Poor electrical conductors
- Brittle (if solid) or gaseous
- Non-lustrous
- Tend to gain electrons

SLIDE 6 *The Periodic Table*

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Atomic Bonding

The Octet Rule: Atoms are most stable with 8 valence electrons — they bond (lose, gain, or share electrons) to reach that stable state.

Metallic Bond

Metal + Metal

Electrons leave individual atoms and form a shared 'electron sea' — flowing freely through the structure.

Effect on material:

Conducts heat & electricity · Bends without fracturing · Atoms slide past each other

← Our primary focus

Ionic Bond

Metal + Non-metal

One atom gives away electrons; the other takes them. Creates oppositely charged ions that attract strongly.

Effect on material:

High melting point · Hard but brittle · Conducts only when molten or dissolved

Covalent Bond

Non-metal + Non-metal

Neither atom gives up electrons — they share them instead, forming strong directional bonds.

Effect on material:

Extremely strong · Poor conductors · Can be very hard (diamond = covalent network)

SLIDE 7 Atomic Bonding

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Crystal Structures — Unit Cells

When metals solidify, atoms arrange into repeating 3D patterns. The unit cell is the smallest repeating building block — like a tile in a tiled floor.

SC	BCC	FCC	HCP
Simple Cubic Packing: 52% Ductility: —	Body-Centered Cubic Packing: 68% Ductility: Low	Face-Centered Cubic Packing: 74% Ductility: High	Hexagonal Close-Packed Packing: 74% Ductility: Med
Atoms at 8 corners only. Teaching example — not found in real engineering metals.	Corners + 1 center atom. Stronger, less ductile. Fe (room temp), W, Cr, Mo, V	Corners + center + face atoms. Most ductile — most slip planes. Al, Cu, Ni, Fe (>912°C), Au, Ag	Same packing as FCC, fewer slip planes → less ductile. Ti, Mg, Zn, Co

SLIDE 9 Crystal Structures — Unit Cells

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From unit cell to bulk metal:

Single repeating building block (~0.3 nm). All atoms perfectly aligned.

Millions of unit cells — all with the same orientation. One crystal domain.

Countless grains packed together, each pointing a different direction.

Grain boundaries block dislocation movement — the mechanism by which deformation spreads through metal. More boundaries = stronger barriers to deformation.

A collage of nine overlapping squares in various shades of blue, green, brown, and purple, arranged in a non-uniform grid pattern. The squares are of different sizes and are layered on top of each other, creating a sense of depth and movement. The colors are muted and earthy, with some squares having thin white borders. The overall composition is abstract and modern.

Grain boundaries = white lines between regions

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Key Takeaways

1 Metallic bonding explains metals

Shared electron sea → electrical and thermal conductivity. Non-directional bonding → ductility. This is what separates metals.

2 Navigate the periodic table

Transition metals (Groups 3–12) are the focus. Elements in the same column share behavior — useful for predicting alloy effects.

3 Structure exists at every scale

Atoms arrange into unit cells → unit cells form grains → grains make bulk metal. Each level determines performance.

4 Know BCC, FCC, and HCP

FCC is most ductile (74%, many slip planes). BCC is stronger but less ductile. HCP has 74% packing but fewer slip planes — less ductile than FCC.

SLIDE 11 Key Takeaways

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Week 1 Resources

- Moniz Ch. 4 — Structure of Metals
- Kalpakjian Ch. 3 — Structure and Manufacturing Properties of Metals

- Crystal Structures Visualization
- Periodic Table of Elements

Mechanical Properties & Behavior

- Stress and strain — the language of mechanical behavior
- Elastic vs. plastic deformation and the yield point
- Crystal defects — point, line, planar, and volume types
- Grain boundaries, Hall-Petch, and diffusion through solid metal

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